

SQL Practical

Consider an organization that maintains information about its **employees, departments, and projects** using the relations

EMPLOYEE (Emp_ID, Emp_Name, Gender, Salary, Dept_ID, Project_ID),
DEPARTMENT (Dept_ID, Dept_Name, Location, Manager_ID), and
PROJECT (Project_ID, Project_Name, Budget, Dept_ID).

Create these three tables in a database and insert the sample data provided in the exercise. Using this database, perform the given queries using **both SQL and Relational Algebra**. For each query, write the appropriate SQL statement and Relational Algebra expression, execute the SQL query to obtain the result, and verify the corresponding Relational Algebra expression using a **Relational Algebra Calculator**. Finally, compare the outputs obtained from SQL and Relational Algebra and observe how **Selection (σ)**, **Projection (π)**, and **Join ($\bowtie$)** operations are used to retrieve the required information.

Emp_ID	Emp_Name	Gender	Salary	Dept_ID	Project_ID
101	Amit Sharma	M	₹75,000	D01	P01
102	Priya Verma	F	₹58,000	D02	P02
103	Rahul Khan	M	₹65,000	D01	P03
104	Neha Singh	F	₹72,000	D03	P01
105	Arjun Patel	M	₹55,000	D02	P04
106	Sneha Gupta	F	₹90,000	D01	P03
107	Rohit Mehta	M	₹62,000	D03	P02
108	Pooja Sharma	F	₹48,000	D02	P04

Dept_ID	Dept_Name	Location	Manager_ID
D01	Research	Bhopal	201
D02	Development	Indore	202
D03	Human Resources	Bhopal	203
D04	Finance	Delhi	204

Project_ID	Project_Name	Budget	Dept_ID
P01	Database Management System	₹15,00,000	D01
P02	E-Commerce Portal	₹8,00,000	D02
P03	Artificial Intelligence	₹25,00,000	D01
P04	Employee Management System	₹12,00,000	D03

Query 1: Employees whose salary is greater than ₹60,000

$$\pi_{Emp_Name, Salary} (\sigma_{Salary > 60000} (EMPLOYEE))$$

```
SELECT Emp_Name, Salary
FROM EMPLOYEE
WHERE Salary > 60000;
```

Emp_ID	Emp_Name	Gender	Salary	Dept_ID	Project_ID
101	Amit Sharma	M	₹75,000	D01	P01
102	Priya Verma	F	₹58,000	D02	P02
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Query 2: Employees working in the 'Research' department

```
SELECT E.Emp_Name
FROM EMPLOYEE E
JOIN DEPARTMENT D
ON E.Dept_ID = D.Dept_ID
WHERE D.Dept_Name = 'Research';
```

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Query 2: Employees working in the 'Research' department

$$\pi_{Emp_Name} (\sigma_{Dept_Name='Research'}$$

$$(EMPLOYEE \bowtie_{EMPLOYEE.Dept_ID=DEPARTMENT.Dept_ID} DEPARTMENT))$$

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Query 3: Projects with a budget greater than ₹10,00,000

```
SELECT Project_Name
FROM PROJECT
WHERE Budget > 1000000;
```

Emp_ID	Emp_Name	Gender	Salary	Dept_ID	Project_ID
101	Amit Sharma	M	₹75,000	D01	P01
102	Priya Verma	F	₹58,000	D02	P02
103	Rahul Khan	M	₹65,000	D01	P03
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Query 3: Projects with a budget greater than ₹10,00,000

$$\pi_{Project_Name} (\sigma_{Budget > 1000000} (PROJECT))$$

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Query 4: Employee names along with their department names

```
SELECT E.Emp_Name, D.Dept_Name
FROM EMPLOYEE E
JOIN DEPARTMENT D
ON E.Dept_ID = D.Dept_ID;
```

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$$\pi_{Emp_Name, Dept_Name}$$

$$(EMPLOYEE \bowtie_{EMPLOYEE.Dept_ID=DEPARTMENT.Dept_ID} DEPARTMENT)$$

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Query 5: Employees assigned to the project 'Database Management System'

```

SELECT E.Emp_Name
FROM EMPLOYEE E
JOIN PROJECT P
ON E.Project_ID = P.Project_ID
WHERE P.Project_Name = 'Database Management System';

```

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$$\pi_{Emp_Name} \left(\sigma_{Project_Name='Database Management System'} \right. \\ \left. (EMPLOYEE \bowtie_{EMPLOYEE.Project_ID=PROJECT.Project_ID} PROJECT) \right)$$

Practice Queries

1. Find the names of all male employees.
2. Find the names and salaries of employees whose salary is less than ₹60,000.
3. Find the names of employees who earn ₹70,000 or more.
4. Find the names of employees who work in the 'Development' department.
5. Find the names of employees who work in departments located in 'Bhopal'.
6. Find the names of employees along with their project names.
7. Find the names of employees working on projects whose budget is greater than ₹10,00,000.
8. Find the names of projects managed by the 'Research' department.
9. Find the names of employees who are working on the 'Artificial Intelligence' project.
10. Find the names, salaries, and department names of employees whose salary is greater than ₹60,000.

Submission

- Screenshots of queries along with output on Relational algebra Calculator
- Single PDF or Doc file